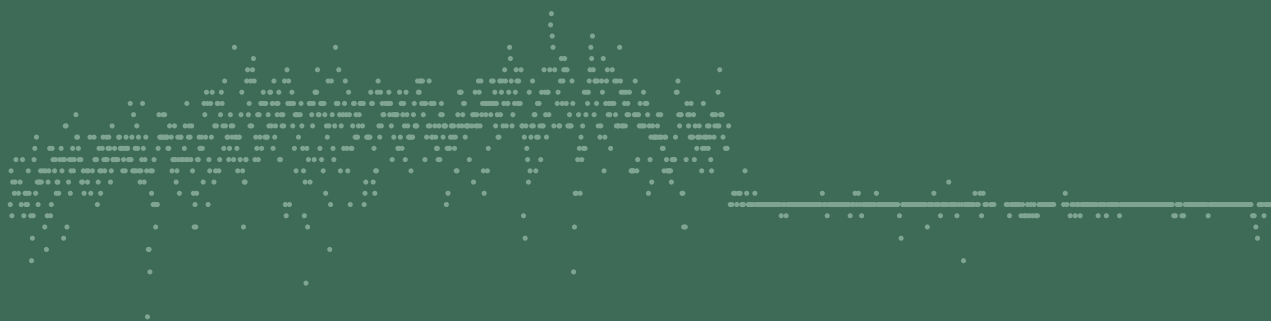




Each dot is one leaflet: its date (left to right, 1997–2026) and how many plants it featured (bottom to top).
The flat line from 2014 is a change of format: leaflets became a fixed walk of 15 stops, published fortnightly.

A REPORT FROM THE CANBERRA BLOOM RECORD

Are the Gardens' plants flowering earlier?

Evidence from 1,147 volunteer flower-walk leaflets written at the Australian National Botanic Gardens, Canberra, from 1997 to 2026.

–5.5 days

change in flowering time per
decade, averaged over all
plants

18–36%

of that change matched by
Canberra's warming

436

plants and cultivars with
enough records to track

22,307

records of a plant in flower on a
given day

How to read this report. The first two pages give the answer in plain language. The sections after that show the evidence, the charts behind it, and how much weight it can bear. A glossary at the end explains the few technical terms. Every figure here comes from the same data as the website, where each leaflet and each plant can be looked up and checked.

Website: <https://rossed.github.io/canberra-bloom-record/>

SUMMARY

The short answer

The plants in the Gardens now flower about **5 days earlier per decade** than in 1997, roughly two weeks over the whole period. Warming explains roughly 18–36% of the shift.

Where the evidence comes from. Every week or two since 1997 a volunteer has walked through the Gardens and written a leaflet naming the plants in flower along the way. The Gardens published these as *In Flower This Week* until August 2016, and the Friends of the ANBG have continued them since as *Flowers, Fruit & Foliage*. In every leaflet the featured plants are printed in bold. That makes it possible to list, for each plant, the dates it was in flower, year after year.

What was found

- **Flowering is earlier.** Across 436 plants seen in at least eight years, the middle of each plant's flowering period moved -5.5 days per decade (95% confidence interval -8.0 to -2.9). Of the 322 plants with a clear flowering season, 215 (67%) moved earlier.
- **The result holds when checked.** It holds when the weekly leaflets of the early years are thinned to match the later fortnightly ones, when long-flowering plants are left out, and within the Gardens' own leaflets alone. Nothing jumps at the 2016 handover to the Friends.
- **Warmth plays a part.** Canberra has warmed by about a third of a degree each decade since 1997, and in warmer years plants flowered 3.4–7.4 days earlier for each degree of extra warmth.
- **But warming explains only part of it.** On these figures, warming accounts for roughly a fifth to a third of the change (18–36%). Rainfall made no clear difference. The rest is probably the Gardens themselves changing (plants maturing, new plantings, watering) and changes in who wrote the leaflets and which paths they walked.
- **Plant families move together with the garden.** Plant families are not shifting in their own distinctive ways: family explains only 6% of the differences between species. Most families are moving earlier along with the garden as a whole, and the wattles are the main exception.

How much weight does this bear?

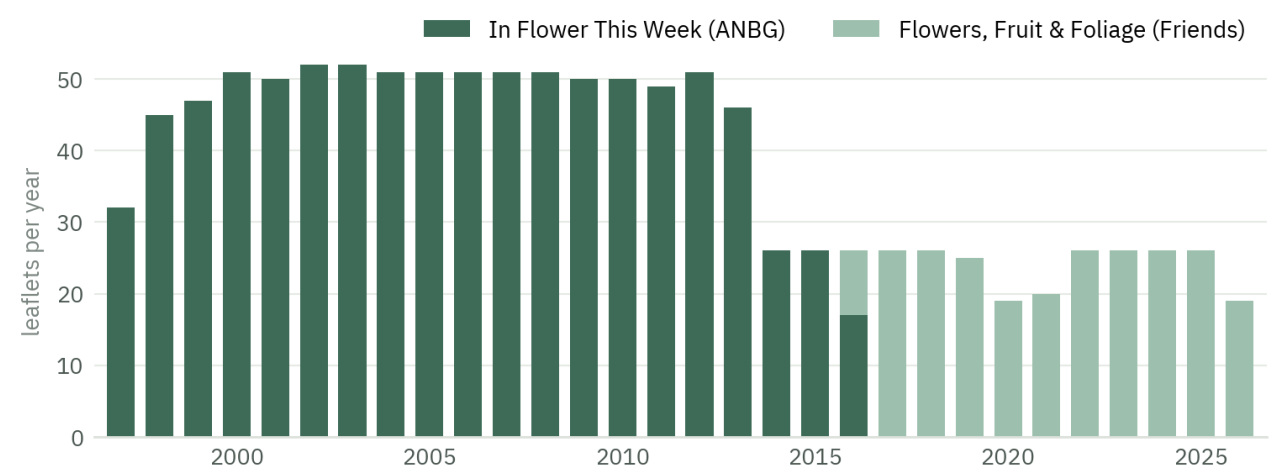
The leaflets were written to guide visitors, not as a scientific survey. A plant only appears when a volunteer chose to walk past it and mention it. The overall pattern across hundreds of plants is fairly robust. The result for any single plant is weak evidence and should be treated as a lead to follow up, not a finding. Read the whole report as a strong hint rather than proof.

1 · THE EVIDENCE

Thirty years of flower-walk leaflets

The report draws on two series of leaflets that follow the same format. Both describe a short walk through the Gardens, stop by stop, with the plants in flower printed in bold.

Series	Published by	Period	Format	Leaflets
<i>In Flower This Week</i>	Australian National Botanic Gardens	Apr 1997 – Aug 2016	Web pages; weekly, fortnightly from 2014	899
<i>Flowers, Fruit & Foliage</i>	Friends of the ANBG	Aug 2016 – Sep 2026	PDF leaflets; fortnightly	248



Leaflets per year. The drop in 2014 is the switch from weekly to fortnightly leaflets. At the same time each leaflet became a fixed walk of 15 stops, down from an average of about 16–23 plants, so later years have about a third as many records. Section 2 explains how the analysis allows for this. 181 leaflets are no longer on their original websites and were recovered from the Internet Archive's copies. There are no leaflets for April–June 2020 or mid-August to October 2021, the periods of COVID-19 restrictions in Canberra.

Weather data

Daily temperatures come from the Bureau of Meteorology's ACORN-SAT record for Canberra Airport, a long-running series that the Bureau adjusts for changes in instruments and location. It currently runs to 2024. For the months since then, the ERA5 global weather reanalysis for the Gardens' location was used, corrected month by month to match the Bureau's record (the two agree closely: $r = 0.95$). Rainfall comes from ERA5 throughout.

2 · METHOD

From leaflets to flowering dates

The process has five steps. Each was automated so it can be rerun and checked.

1

Collect every leaflet

All 899 Gardens web pages and 248 Friends PDFs listed on the organisations' websites were downloaded. Where a link had gone dead, the Internet Archive's copy was used. Some early archived PDFs had been cut short at 1 MB, and for those the most complete copy was chosen.

2

Pick out the plants

A program read each leaflet and kept the words printed in bold that look like plant names. Italic Latin names and cultivar names in quotes, such as *Grevillea* 'Lady O', were joined up. The Friends' PDFs are laid out in two columns, so they were read stop by stop. Ten leaflets from 2019–20 stored their text in a scrambled form and were decoded. This gave 22,307 records of a plant in flower on a given date.

3

Tidy the names

Botanists rename plants, and leaflets sometimes misspell them. Every name was checked against the Atlas of Living Australia, which gives the currently accepted name. For example, *Bracteantha bracteata* is now *Xerochrysum bracteatum*. 83% of records were matched to a species and 99.7% to at least a plant family. 1,078 records were printed under an older name.

4

Work out flowering time

For each plant and each year, the middle of the dates it was mentioned is its **mid-flowering date**. Each plant's year starts at its quietest time, so summer bloomers aren't split at New Year. A year only counts if the leaflets covered that plant's season with no gap longer than 28 days. Plants needed at least 8 such years spread over 10+ years, which left 436 plants and cultivars.

5

Compare with the weather

For each plant and year, the average temperature in the weeks before its usual flowering start was compared with how early or late it flowered that year. Garden-wide results pool all plants, and the uncertainty allows for plants in the same year sharing the same weather, volunteer and route.

Why the middle of flowering, not the first sighting?

The most obvious measure would be the first date each year a plant appears. But in 2014 the leaflets went from weekly to fortnightly, and with half as many leaflets a plant tends to be spotted about a week later, even if nothing has changed. Taken at face value, first sightings show only –1.4 days per decade. When the early weekly years are thinned to fortnightly so both periods are sampled alike, first sightings show –4.1 days per decade (95% CI –6.7 to –1.6), close to the mid-flowering result. The mid-flowering date isn't thrown off by the sampling change, so it is the main measure in this report.

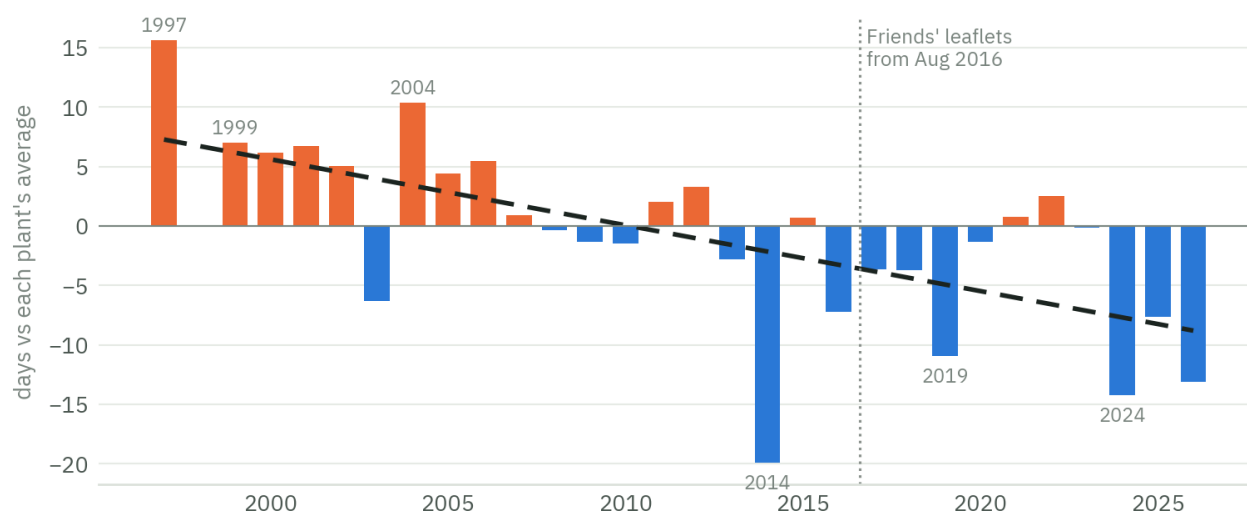
Checking the extraction

The website's Leaflets page shows every leaflet next to the plants taken from it, and flags 56 leaflets for a closer look. A Check accuracy page lets reviewers compare a random sample of leaflets with the originals. Reviewer results will appear here when available. On the website, each dot on a plant's chart links to its leaflet.

3 · FINDINGS

The whole garden is flowering earlier

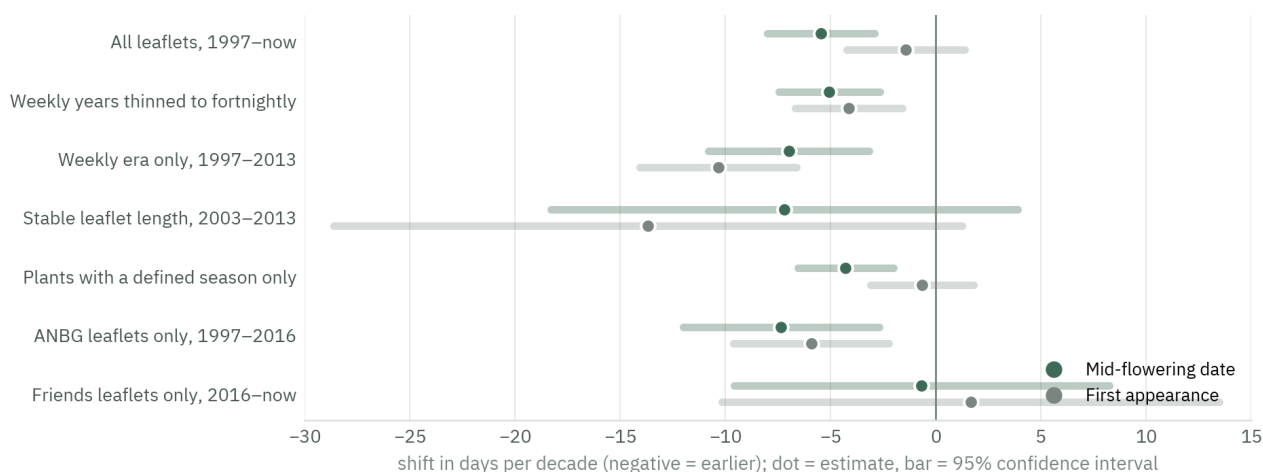
The chart below combines all 436 plants. For each plant and year, it takes how many days earlier or later than that plant's usual mid-flowering date it flowered, then averages across plants. Bars below zero are years when the garden as a whole flowered early.



Garden-wide flowering index by year (years with at least 30 plant records). The dashed line is the trend: -5.5 days per decade (95% CI -8.0 to -2.9). The earliest years were 2014, 2019 and 2024, and the latest 1997, 1999 and 2004.

Is the result robust?

Changes in how the leaflets were produced could create a false trend. So the analysis was repeated on subsets of the data that each remove one possible distortion. Green shows the main measure (mid-flowering date) and grey the first-sighting measure. A result is solid when the green dots stay left of zero and their bars don't cross it.

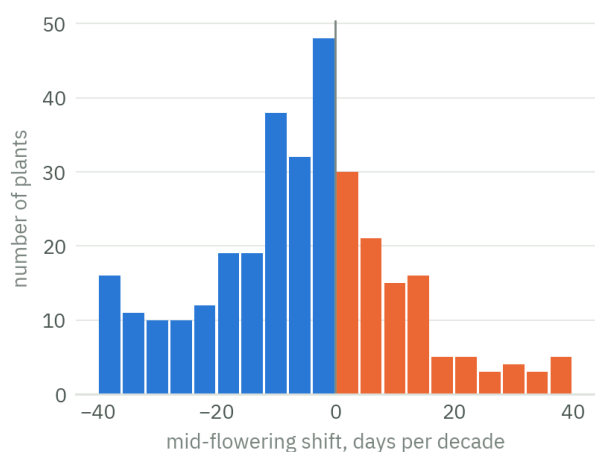


Each row repeats the garden-wide calculation on a subset. Only the Friends-only row, with just 88 plants over ten years, is too small to say anything. After allowing for the trend, the Friends' leaflets sit $+3.4$ days from the Gardens' ones (p 0.408), so the handover did not create the result.

4 · WHICH PLANTS

Which plants are shifting

Most plants moved earlier, but by different amounts, and a few moved later. The left chart shows the spread across the 322 plants with a clear flowering season. The box on the right summarises whether plant families shift together.



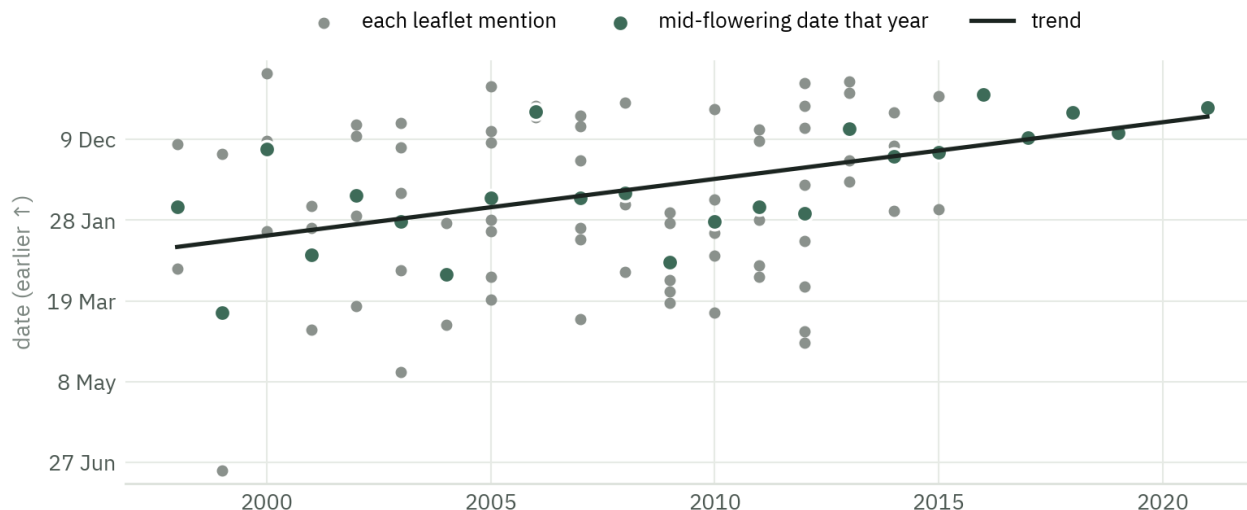
Do families shift together?

Mostly no. Family explains only 6% of the differences between species, no more than random groupings would (p 0.359). Species within a family vary widely. The detail follows the species tables.

215 of 322 plants (67%) moved earlier. If nothing were changing, about half would by chance.

One plant in detail

Hibbertia pedunculata (Stalked Guinea-flower) has the strongest evidence of an earlier shift among wild species. Each grey dot is one leaflet that mentioned it. The green dots mark the middle of each year's mentions.



Mid-flowering moved -35.1 days per decade over 23 well-covered years (1998–2021). The earliest dates are at the top.

4 · WHICH PLANTS (CONTINUED)

Plants with the clearest shifts

These tables list the plants whose change is least likely to be chance, measured by the p value. With 436 plants tested, a few will look significant by luck. The q value corrects for that: q below 0.1 means the shift is unlikely to be a false alarm. 6 plants moved earlier and 1 later with q below 0.1. Treat the rest as leads.

Moving earlier

Plant	Family	Usually starts	Years	Shift (days/decade)	p	q
<i>Hibbertia pedunculata</i> Stalked Guinea-flower	Dilleniaceae	30 Nov	23	−35.1	<0.001	0.07
<i>Melaleuca linariifolia</i> Snow-in Summer	Myrtaceae	24 Dec	9	−52.9	<0.001	0.07
<i>Lambertia formosa</i> Mountain Devil	Proteaceae	25 Mar	13	−47.1	<0.001	0.07
<i>Cuttsia viburnea</i> Elderberry	Rousseaceae	4 Dec	20	−7.2	0.001	0.07
<i>Grevillea corrugata</i>	Proteaceae	31 Aug	11	−36.7	0.001	0.07
<i>Grevillea johnsonii</i> Johnson's Grevillea	Proteaceae	29 Oct	9	−45.6	0.003	0.12
<i>Hybanthus floribundus</i> Shrub Violet	Violaceae	16 Jul	12	−47.0	0.004	0.14
<i>Banksia aquilonia</i> Northern Banksia	Proteaceae	29 Apr	11	−33.2	0.006	0.14
<i>Banksia neoanglica</i> New England Banksia	Proteaceae	10 May	16	−36.6	0.006	0.14
<i>Verticordia plumosa</i> Plumed Featherflower	Myrtaceae	30 Nov	14	−10.7	0.007	0.15
<i>Callistemon brachyandrus</i> Prickly Bottlebrush	Myrtaceae	24 Jan	14	−13.4	0.010	0.20
<i>Eremophila subfloccosa</i> Dense-felted Eremophila	Scrophulariaceae	5 Jun	8	−66.7	0.012	0.22
<i>Banksia</i> 'Red Clusters' Heath-leaved Banksia	Proteaceae	17 May	10	−32.4	0.017	0.28
<i>Hakea purpurea</i>	Proteaceae	21 Jul	17	−25.6	0.018	0.28
<i>Persoonia acerosa</i> Needle Geebung	Proteaceae	20 Jan	12	−33.1	0.019	0.28
<i>Banksia blechnifolia</i>	Proteaceae	21 Sep	15	−89.8	0.021	0.28
<i>Leptospermum squarrosum</i> Peach Blossom Tea Tree	Myrtaceae	17 Apr	15	−33.1	0.023	0.30
<i>Grevillea lavandulacea</i> Lavender Grevillea	Proteaceae	27 Aug	15	−18.9	0.023	0.30

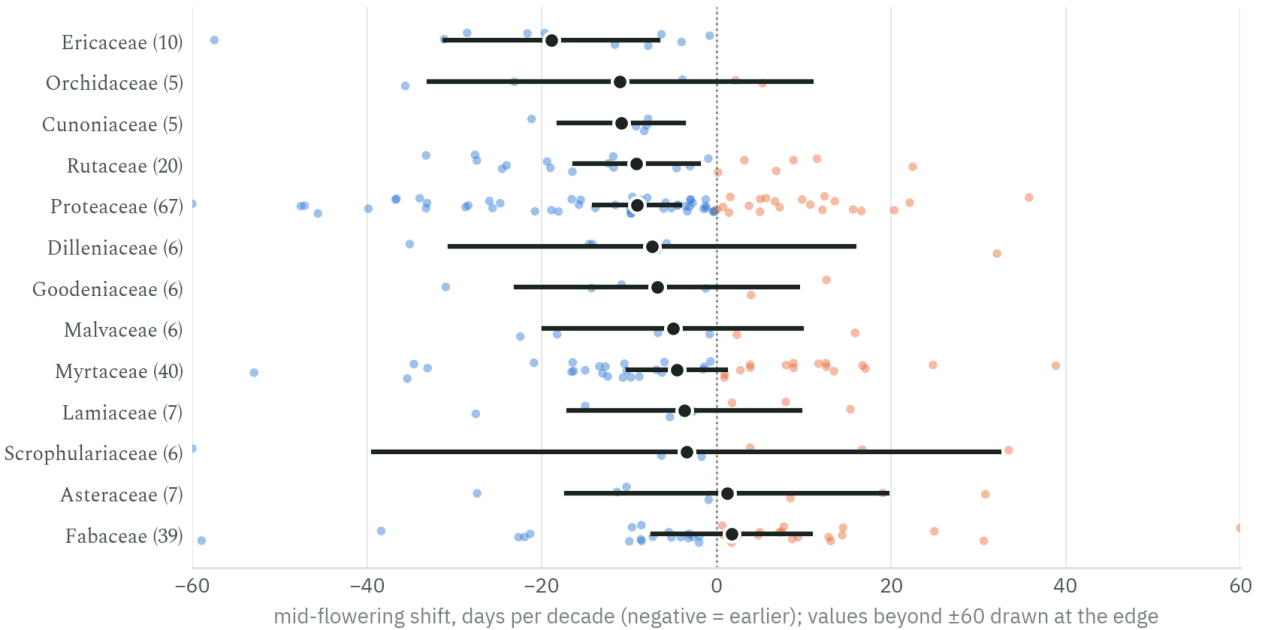
Moving later

Plant	Family	Usually starts	Years	Shift (days/decade)	p	q
<i>Platylobium formosum</i> Handsome Flat-pea	Fabaceae	1 Dec	8	+141.1	0.003	0.12
<i>Eriostemon australasius</i> Waxflower	Rutaceae	17 Oct	11	+22.5	0.004	0.14
<i>Grevillea</i> ‘Poorinda Royal Mantle’	Proteaceae	20 Nov	24	+22.9	0.019	0.28
<i>Acacia beckleri</i> Barrier Range Wattle	Fabaceae	10 May	9	+14.4	0.020	0.28
<i>Homoranthus flavescens</i> Mouse-and-honey Bush	Myrtaceae	1 Sep	18	+24.8	0.032	0.36
<i>Telopea</i> ‘Canberry Gem’ Waratahs	Proteaceae	26 Oct	12	+29.3	0.041	0.37
<i>Acacia</i> ‘Green Mist’ Narrow-leaf Bower-wattle	Fabaceae	6 Sep	13	+21.3	0.061	0.42
<i>Acacia terminalis</i> Sunshine Wattle	Fabaceae	4 Jul	14	+30.7	0.064	0.42

Long-flowering plants (in flower for more than about five months of the year) are left out. Their dates mostly reflect which weeks a volunteer happened to pass, not when they flowered. The full list of plants, with search and charts, is on the website's Findings page.

Do plant families shift together?

Each family's figure is simply the average of its species. The chart shows every species as a dot, so the spread is visible. Species within a family typically differ by ± 21 days per decade. About ± 15 of that is noise in the leaflet records, leaving roughly ± 14 of real difference between species.



Wild species with a defined season, in families with at least 3 such species (224 species). Small dots are species (blue earlier, orange later). The black dot is the family average and the line its 95% confidence interval. When the line doesn't cross zero, the family as a whole is shifting.

The verdict. Family explains only 6% of the differences between species. Random groupings of the same species do as well 36% of the time (permutation test p 0.359), so families are not moving in their own distinctive ways. Most drift earlier with the garden as a whole. 4 families are clearly earlier even after allowing for testing 13 families: Ericaceae, Cunoniaceae, Rutaceae and Proteaceae. The pea and wattle family (Fabaceae) is the exception at +1.8 days per decade, with *Acacia* at -0.3 . Genera differ slightly more than families (p 0.046), a hint worth following up rather than a finding.

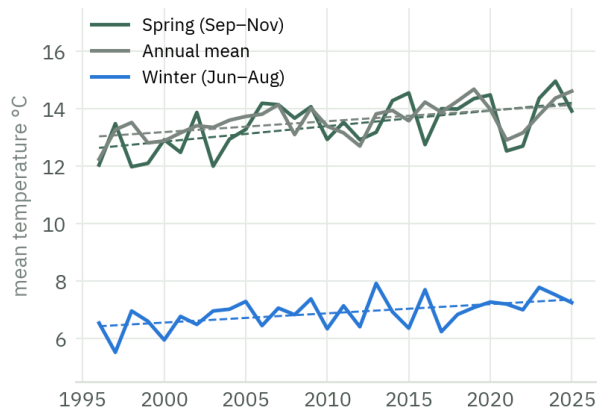
Family	Species	Average	95% CI	Spread	Earlier	q	Verdict
Ericaceae	10	-18.9	-31 to -7	±17	10/10	0.04	clearly earlier
Orchidaceae	5	-11.0	-33 to +11	±18	3/5	0.51	no clear shift
Cunoniaceae	5	-10.9	-18 to -4	±6	5/5	0.05	clearly earlier
Rutaceae	20	-9.1	-16 to -2	±15	14/20	0.05	clearly earlier
Proteaceae	67	-9.1	-14 to -4	±20	47/67	0.01	clearly earlier
Dilleniaceae	6	-7.4	-31 to +16	±22	5/6	0.65	no clear shift
Goodeniaceae	6	-6.8	-23 to +9	±15	4/6	0.61	no clear shift
Malvaceae	6	-5.0	-20 to +10	±14	4/6	0.65	no clear shift
Myrtaceae	40	-4.5	-10 to +1	±17	24/40	0.29	no clear shift
Lamiaceae	7	-3.6	-17 to +10	±14	4/7	0.69	no clear shift
Scrophulariaceae	6	-3.4	-39 to +32	±34	3/6	0.88	no clear shift
Asteraceae	7	+1.2	-17 to +20	±20	4/7	0.88	no clear shift
Fabaceae	39	+1.8	-7 to +11	±28	19/39	0.82	no clear shift

Shifts in days per decade. Spread is the standard deviation across the family's species. q is corrected for testing all families at once.

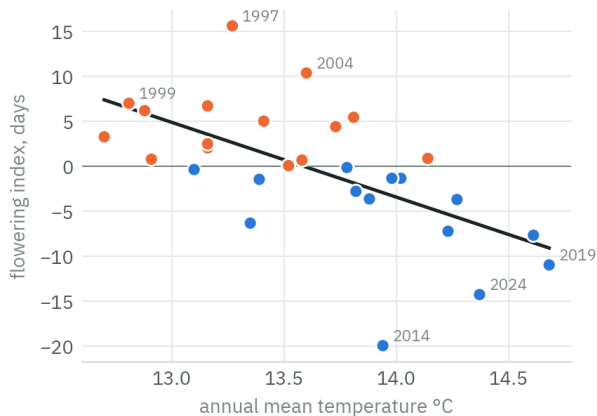
5 · WHY

Is it the warming?

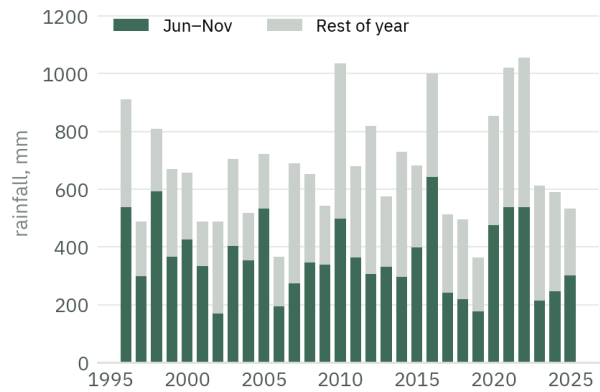
Canberra has warmed over the period, especially in winter and spring. Warmer years clearly flower earlier. But the warming is modest, and on its own it accounts for only part of the change.



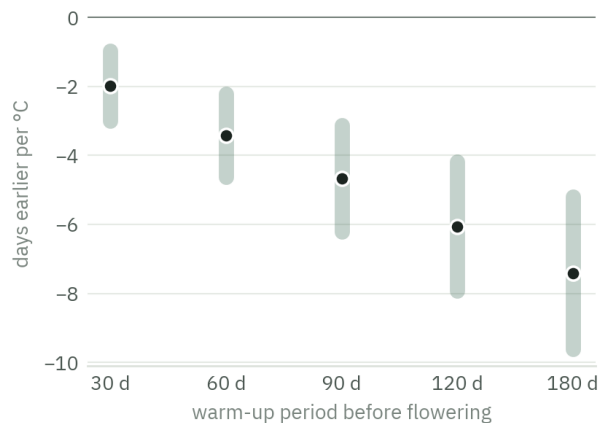
Canberra seasonal mean temperature with trend lines. Per decade, 1997–2025: annual +0.32 °C, winter +0.33, spring +0.50.



Each dot is a year. Warmer years flowered earlier ($r = -0.63$). With the long-term trend removed from both, the link remains ($r = -0.45$, $p = 0.013$).



Rainfall. The Millennium Drought lasted until 2009. 2010 and the La Niña years of 2021–22 were very wet. Unlike temperature, rainfall shows no clear link with flowering time.



Days earlier per °C, depending on how long a warm-up before flowering is measured. Longer periods show stronger effects, up to 7.4 days per °C.

5 · WHY (CONTINUED)

Putting it together

- **How much can warming explain?** Plants flower 3.4–7.4 days earlier per °C, and the relevant periods warmed about 0.27–0.29 °C per decade. That predicts –1.0 to –2.0 days per decade, against –5.5 observed, or about 18–36%.
- **The rest isn't explained by weather.** With temperature, rainfall and time in one model, a shift of –4.6 days per decade remains. This is a lower bound on warming's share, because a fixed warm-up period only approximates when each plant responds.
- **Rainfall makes no clear difference.** Neither wetter years ($r = +0.08$ after removing trends) nor rain before each plant's own season shows a consistent effect on flowering time.
- **Other likely causes.** The garden is maturing and watered, which can make plants less tied to the weather. New plantings and cultivars are often chosen because they flower early or for long periods. The volunteers and their routes changed over time. The leaflets can't separate these effects.

6 · LIMITATIONS

What this can and can't tell us

- **The leaflets are not a survey.** A plant is recorded only when a volunteer chose to feature it. Its absence in a given week means little.
- **Single plants are weak evidence.** With hundreds of plants tested, some will show large shifts by chance. Only the q values guard against that, and only a handful of plants pass.
- **The way the leaflets were made changed.** The switch to fortnightly leaflets in 2014, longer leaflets in the 2000s, new writers and new routes could all nudge the results. The checks in section 3 remove each of these in turn, and the result held.
- **Automated reading makes mistakes.** The website shows every extracted plant beside its leaflet so errors can be found, and a checking page measures the error rate.
- **Weather is measured at the airport.** Canberra Airport is about 8 km from the Gardens. Year-to-year changes track well, but the Gardens' own conditions differ, and it is irrigated. Temperatures after 2024 come from a weather model, not the Bureau's adjusted record.
- **Correlation is not proof of cause.** Warm years flowering early fits a warming effect, but the leaflets alone can't rule out other explanations.

GLOSSARY AND CREDITS

Glossary

Term	Meaning
Mid-flowering date	The middle of the dates a plant was mentioned in a given year. The main measure in this report.
First appearance	The first date in a year a plant was mentioned. Affected by how often leaflets were published.
Days per decade	How much a date moved, on average, for every ten years that passed. Negative means earlier.
95% confidence interval	The range the true value probably lies in. If it doesn't include zero, the change is unlikely to be chance.
p value	How likely a result this large would be if nothing were really changing. Below 0.05 is usually called significant.
q value	A p value adjusted for testing many plants at once. Below 0.1 means the result is unlikely to be a false alarm.
r (correlation)	How closely two things move together, from -1 to $+1$. Zero means no relationship.
Taxon, cultivar	A taxon is any named plant group, here usually a species. A cultivar is a named garden variety, written in quotes.
ACORN-SAT, ERA5	The Bureau of Meteorology's adjusted temperature record, and a global weather reanalysis used to fill recent months and rainfall.

Credits and sources

Leaflets written by the volunteers of the Australian National Botanic Gardens and the Friends of the ANBG. Short excerpts are used for analysis only. Plant names: Atlas of Living Australia. Temperature: Bureau of Meteorology ACORN-SAT (CC BY). Rainfall and recent temperature: Copernicus ERA5 via Open-Meteo (CC BY 4.0). The website, code and processed data are at <https://rossed.github.io/canberra-bloom-record/> and <https://github.com/Rossed/canberra-bloom-record>. Report generated 27 September 2026.